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CANDIDATE NUMBER

2022 Trial HSC Examination

Form VI Mathematics Extension 2

Wednesday 10th August 2022

8:40am

General Instructions

- Reading time — 10 minutes
- Working time — 3 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Total Marks: 100

Section I (10 marks) Questions 1 – 10

- This section is multiple-choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (90 marks) Questions 11 – 16

- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Write your candidate number on this page, on each booklet and on the multiple choice sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.
- Place everything inside this question booklet.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 6 booklets per boy
- Candidature: 78 pupils

Writer: WJM

Section I

Questions in this section are multiple-choice.

Record the single best answer for each question on the provided answer sheet.

1. What is the value of $(-1 + i)^2$?

(A) -2
(B) 2
(C) $-2i$
(D) $2i$

2. Consider the statement:

If Bob eats an apple every day, then he does not have to visit a doctor.

What is the **converse** of this statement?

(A) If Bob does not eat an apple every day, then he has to visit a doctor.
(B) If Bob eats an apple every day, then he has to visit a doctor.
(C) If Bob has to visit a doctor, then he does not eat an apple every day.
(D) If Bob does not have to visit a doctor, then he eats an apple every day.

3. The two lines $r_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $r_2 = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ intersect at the point $(2, 5, -1)$.

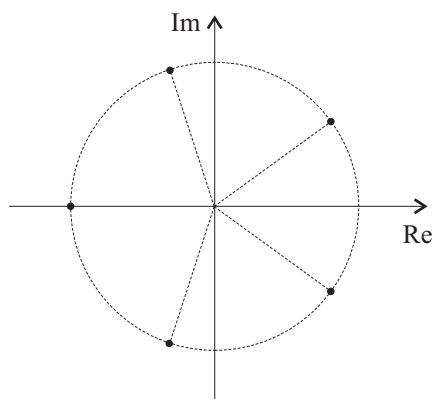
What is the approximate size of the angle between r_1 and r_2 ?

(A) 50°
(B) 82°
(C) 97°
(D) 140°

4. Which of the following integrals is equivalent to $\int \sqrt{\frac{1+x}{1-x}} dx$?

(A) $\int \sqrt{1-x^2} dx$
(B) $\int \frac{1+x}{\sqrt{1-x^2}} dx$
(C) $\int \frac{1+x}{1-x} dx$
(D) $\int \left(1 + \frac{2x}{\sqrt{1-x^2}}\right) dx$

5.



The diagram above shows the solutions to $z^5 + a = 0$. Which of the following could be the value of a ?

- (A) 2
- (B) -3
- (C) $4i$
- (D) $-5i$

6. Euler's *sum of powers* conjecture states that for $a_1, a_2, \dots, a_n, b \in \mathbf{Z}$:

If $\exists k \in \mathbf{Z}^+$ such that $a_1^k + a_2^k + \dots + a_n^k = b^k$, then $n \geq k$.

Given that each of the following statements is true, which statement is a counterexample to this conjecture?

- (A) $27^2 + 72^2 + 96^2 = 123^2$
- (B) $133^4 + 134^4 - 59^4 = 158^4$
- (C) $27^5 + 84^5 + 110^5 + 133^5 = 144^5$
- (D) $955^4 + 1770^4 + 2634^4 + 5400^4 = 5491^4$

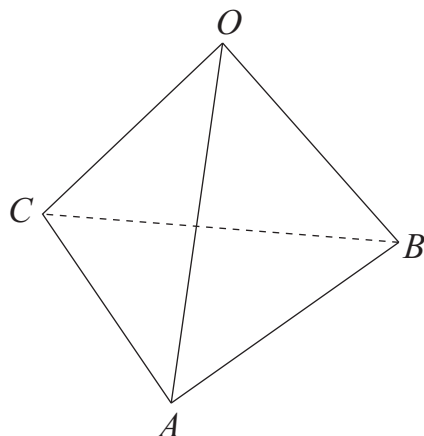
7. The velocity of an object in metres per second is defined by $\dot{x} = 6 \cos \pi t$. What distance does the object travel from $t = 0$ to $t = 1$?

- (A) $\frac{6}{\pi}$ metres
- (B) $\frac{12}{\pi}$ metres
- (C) 6 metres
- (D) 12 metres

8. The complex number z is defined such that $|z| = |z - 1 - i|$. Which of the following describes the set of values $\text{Arg}(z)$ can take?

- (A) $-\pi < \text{Arg}(z) \leq \pi$
- (B) $\text{Arg}(z) = -\frac{3\pi}{4}$ or $\text{Arg}(z) = \frac{\pi}{4}$
- (C) $-\frac{\pi}{2} < \text{Arg}(z) < \frac{\pi}{2}$
- (D) $-\frac{\pi}{4} < \text{Arg}(z) < \frac{3\pi}{4}$

9.



The diagram above shows the tetrahedron $OABC$. That is, $OABC$ is a triangular pyramid where each face is an equilateral triangle. Each edge of the pyramid has length 1 unit.

Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, and $\overrightarrow{OC} = \underline{c}$.

What is the value of $|\text{proj}_{\underline{a}}(\text{proj}_{\underline{c}}(\text{proj}_{\underline{b}}\underline{a}))|$?

- (A) $\frac{1}{8}$
- (B) $\frac{1}{4}$
- (C) 1
- (D) 4

10. Consider the following proof:

For $a, b, c \in \mathbf{Z}^+$, $\exists p, q, r \in \mathbf{Z}^+$ such that $a = 3p + 1$, $b = 3q + 1$, $c = 3r + 1$.

$$\begin{aligned}
 abc &= (3p + 1)(3q + 1)(3r + 1) \\
 &= 27pqr + 9pq + 9pr + 9qr + 3p + 3q + 3r + 1 \\
 &= 3(9pqr + 3pq + 3pr + 3qr + p + q + r) + 1 \\
 &= 3k + 1, \quad k \in \mathbf{Z}^+
 \end{aligned}$$

Which of the following can be deduced from the proof above?

- (A) If the product of three integers is one more than a multiple of 3, then the three integers must all be one more than a multiple of 3.
- (B) If the product of three integers is one more than a multiple of 3, then at least one of the three integers must be one more than a multiple of 3.
- (C) If the product of three integers is two more than a multiple of 3, then none of the three integers can be one more than a multiple of 3.
- (D) If the product of three integers is two more than a multiple of 3, then at least one of the three integers must not be one more than a multiple of 3.

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (16 marks) Start a new answer booklet.

Marks

(a) Given the two complex numbers $z = 1 + \sqrt{3}i$ and $w = 3e^{i\frac{2\pi}{3}}$,

(i) Express z in the form $re^{i\theta}$.

2

(ii) Find zw , fully simplifying your answer.

2

(b) Find:

(i) $\int \frac{\sec^2 x}{1 + 2 \tan x} dx$

1

(ii) $\int xe^x dx$

2

(iii) $\int \cos^3 x dx$

2

(c) Solve $z^2 - 2iz - 5 = 0$.

2

(d) Consider the statement: $\forall x \in \mathbf{R}, \exists y \in \mathbf{R}$ such that $y^2 = x^3$.

1

Determine if this statement is true or false. Justify your answer.

(e) Given the complex numbers $z = 1 + 5i$ and $w = 3 - 2i$, find the real numbers a and b such that $\frac{z}{w} = a + ib$.

2

(f) Consider the line $\mathcal{L} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. The point $A(x, y)$ lies on the line, whereas the point $B(-1, 3)$ does not.

2

If O is the origin, find the coordinates of A if $\overrightarrow{OA} \perp \overrightarrow{OB}$.

QUESTION TWELVE (15 marks) Start a new answer booklet.**Marks**(a) For two vectors $\underline{a}$ and $\underline{b}$, it is known that $|\underline{a}| = 5$, $|\underline{b}| = 3$, and $\underline{a} \cdot \underline{b} = 9$.(i) Show that $\underline{b} \cdot (\underline{a} - \underline{b}) = 0$.

1

(ii) Find $|\underline{a} - \underline{b}|$.

2

(iii) If $\underline{a} = \overrightarrow{OA}$ and $\underline{b} = \overrightarrow{OB}$, find the area of $\triangle OAB$.

1

(b) (i) Prove that if x and y are positive real numbers, then $\frac{x+y}{2} \geq \sqrt{xy}$.

1

(ii) The numbers $a_1, a_2, a_3, \dots, a_n$ are positive and real, and $a_1 a_2 a_3 \dots a_n = 1$.

2

Show that $(1 + a_1)(1 + a_2)(1 + a_3) \dots (1 + a_n) \geq 2^n$.(c) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta}$.

4

(d) Let $I_n = \int_0^1 \frac{x^n}{x+1} dx$ for $n \geq 0$.(i) Use algebraic manipulation to show that $I_n = \frac{1}{n} - I_{n-1}$ for $n \geq 1$.

2

(ii) Hence find I_2 .

2

QUESTION THIRTEEN (14 marks)

Start a new answer booklet.

Marks

- (a) The polynomial $P(z) = z^4 - 10z^3 + cz^2 + dz + 169$ has two zeroes $\alpha = a + ib$ and $\beta = b + ia$, where $a, b, c, d \in \mathbf{R}$ and $a \neq b$.
- (i) Express the other two zeroes of $P(z)$ in terms of a and b , justifying your answer.

1

- (ii) Find the values of a and b , given that they are both integers.

2

- (b) A particle moves in a straight line with initial displacement $x = 0$. The velocity of the particle is given by $v = 2e^{-\frac{x}{2}}(x+1)^2$, where velocity is in metres per second.
- (i) Show that the acceleration of the particle is given by $a = 2e^{-x}(x+1)^3(3-x)$.

2

- (ii) Hence find the displacement for which the maximum velocity of the particle will occur, justifying your answer.

2

- (c) (i) Use De Moivre's theorem to show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$.

3

- (ii) Consider the equation $x^4 - 4x^2 + 1 = 0$. Use the substitution $x = 2\cos\theta$ to show that the roots are $2\cos\frac{\pi}{12}$, $2\cos\frac{5\pi}{12}$, $2\cos\frac{7\pi}{12}$, and $2\cos\frac{11\pi}{12}$.

2

- (iii) Hence show that $\cos^2\frac{\pi}{12} + \cos^2\frac{5\pi}{12} + \cos^2\frac{7\pi}{12} + \cos^2\frac{11\pi}{12} = 2$.

2

QUESTION FOURTEEN (14 marks)

Start a new answer booklet.

Marks

- (a) Three points $A(-2, 1, 7)$, $B(-1, 3, 5)$, and $C(-4, 7, 3)$ lie in three-dimensional space. The points A and B lie on the line l .

(i) Express $\overrightarrow{AB}$ and $\overrightarrow{AC}$ as column vectors.

1

(ii) Find the shortest distance from C to the line l .

3

- (b) For constants p , A , and B , $\frac{1}{p^2 - x^2} = \frac{A}{p + x} + \frac{B}{p - x}$.

(i) Find expressions for A and B in term of p .

2

(ii) Hence or otherwise, show that for some constant C ,

1

$$\int \frac{1}{p^2 - x^2} dx = \frac{1}{2p} \ln \left| \frac{p+x}{p-x} \right| + C.$$

- (c) A physicist drops an object with mass m kg from the top of a cliff. As the object falls, it experiences a force due to gravity of magnitude $10m$ Newtons and a resistive force of magnitude mkv^2 Newtons, where v is the velocity of the object in metres per second and k is a positive constant.

The vertical displacement of the object y metres from where it is dropped satisfies

$$m\ddot{y} = 10m - mkv^2,$$

where the downwards direction is taken as positive.

- (i) Let V be the terminal velocity of the object. Find an expression for V in terms of k , and hence show that $\frac{dv}{dt} = k(V^2 - v^2)$.

2

(ii) Use part (b)(ii) to show that

3

$$v = V \times \frac{e^{2Vkt} - 1}{e^{2Vkt} + 1}.$$

- (iii) The physicist finds that it takes the object 1 second to reach half of its terminal velocity. Find the value of k . Give your answer correct to two decimal places.

2

QUESTION FIFTEEN (16 marks)

Start a new answer booklet.

Marks

(a) Sketch the region on the Argand diagram where $\operatorname{Im} \left(\frac{z + \bar{z}}{z - \bar{z}} \right) \geq 1$. 3

(b) (i) Write down the roots of the equation $z^7 = 1$. 2

(ii) Hence show that 2

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = (z^2 - 2z \cos \frac{2\pi}{7} + 1) (z^2 - 2z \cos \frac{4\pi}{7} + 1) (z^2 - 2z \cos \frac{6\pi}{7} + 1).$$

(iii) Deduce that $\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{8}$. 2

(c) Use mathematical induction to show that $\left(2 - \frac{1}{n}\right)^n > n$ for all integers $n \geq 2$. 3

(d) (i) The function $f(x)$ is continuous and differentiable for all real x . 2

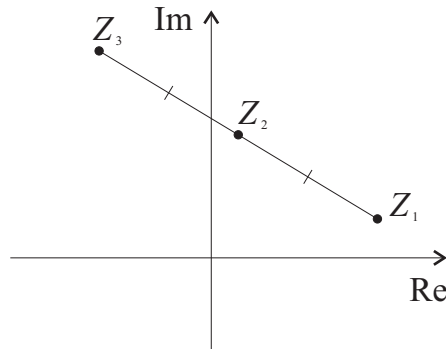
Use the substitution $u = a + b - x$ to show that

$$\int_a^b \frac{f(x)}{f(a+b-x) + f(x)} dx = \frac{b-a}{2}.$$

(ii) Hence use the substitution $u = x^2$ to evaluate $\int_1^3 \frac{x^2}{\sqrt{10-x^2}+x} dx$. 2

QUESTION SIXTEEN (15 marks) Start a new answer booklet.**Marks**

(a)



The diagram above shows the collinear points Z_1 , Z_2 , and Z_3 , which represent the complex numbers z_1 , z_2 , and z_3 respectively. The midpoint of the interval between Z_1 and Z_3 is Z_2 , and it is also known that $z_3 = z_1 z_2$.

(i) Show that $z_2 = \frac{z_1}{2 - z_1}$.

1

(ii) Consider the scenario where z_2 is purely imaginary.

(α) Show that Z_1 will always lie on a particular circle. Find the centre and radius of this circle.

2

(β) Show that the line through Z_1 , Z_2 and Z_3 is a tangent to the circle mentioned in part (ii)(α).

2

(b) (i) By considering the dot product of two vectors, show that for any $a_k, b_k \in \mathbf{R}$,

2

$$(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \geq (a_1 b_1 + a_2 b_2 + a_3 b_3)^2.$$

(ii) Given that x , y , and z are non-negative and $x + y + z = 3$, find the minimum possible value of $x^2 + 4y^2 + z^2$.

2

(c) A function $f(x)$ is continuous and positive in its natural domain, which includes the interval $[0, 3]$.

It is known that $f(3) = 5$, and the function also has the following properties:

- $\int_0^3 (f(x))^2 dx = 81$
- $\int_0^3 (f'(x)f(x))^2 dx = 1$

(i) Show that $\int_0^3 x f'(x) f(x) dx = -3$

3

(ii) By considering $(f'(x)f(x) + kx)^2$, or otherwise, find $f(0)$.

3

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Wednesday 10th August 2022

- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

B L A N K P A G E

Mathematics Ext. 2 Trial - 2022 Solutions

Multiple Choice

$$\textcircled{1} \quad (-1+i)^2 = 1 - 2i + i^2 \\ = -2i$$

$\textcircled{C}$

$\textcircled{2} \quad \textcircled{D}$

$$\textcircled{3} \quad \underline{a} \cdot \underline{b} = |\underline{a}| \times |\underline{b}| \times \cos \theta \quad |\underline{a}| = \sqrt{1^2 + 2^2 + (-1)^2} \\ \cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| \times |\underline{b}|} \quad = \sqrt{6} \\ |\underline{b}| = \sqrt{0^2 + 1^2 + 3^2} \\ = \sqrt{10}$$

$$\underline{a} \cdot \underline{b} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \\ = 0 + 2 - 3 \\ = -1$$

$$\therefore \theta = \cos^{-1} \left(\frac{-1}{\sqrt{6} \times \sqrt{10}} \right) \\ \approx 97^\circ \quad \textcircled{C}$$

$$\textcircled{4} \quad \sqrt{\frac{1+x}{1-x}} = \frac{1+x}{\sqrt{1-x} \sqrt{1+x}} \\ = \frac{1+x}{\sqrt{1-x^2}}$$

$\textcircled{B}$

$\textcircled{5}$ One solution is negative and real.
 $\therefore z^5$ is negative and real
 $\therefore a$ is positive and real

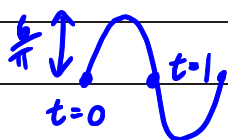
$\textcircled{A}$

$$\textcircled{6} \quad \ln 27^5 + 84^5 + 110^5 + 133^5 = 144^5, \quad n=4 \text{ and } k=5 \\ \text{so } n < k$$

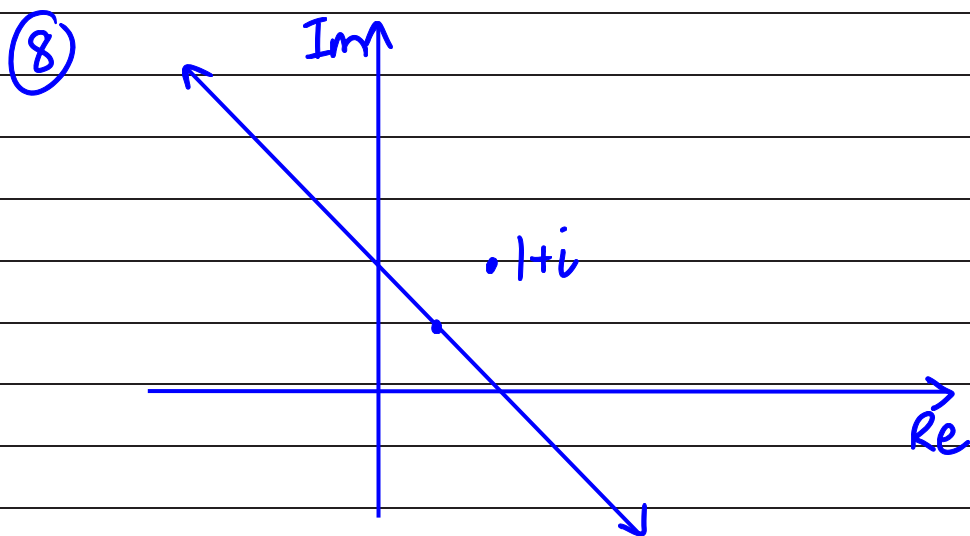
$\therefore \textcircled{C}$

$$\textcircled{7} \quad x = \frac{6}{\pi} \sin \pi t \quad (\text{letting } x=0 \text{ be centre of motion})$$

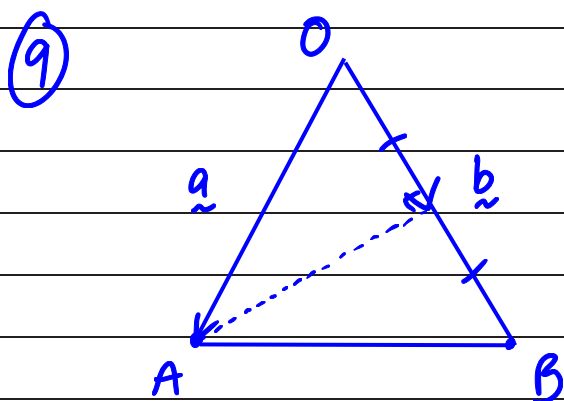
$$\text{Period} = \frac{2\pi}{\pi} \\ = 2$$



$$2 \times \frac{6}{\pi} = \frac{12}{\pi} \Rightarrow \textcircled{B}$$



For any z on this line, $-\frac{\pi}{4} < \text{Arg}(z) < \frac{3\pi}{4}$
 $\therefore$ (D)



$$\text{proj}_{\underline{b}} \underline{a} = \frac{1}{2} \underline{b}$$

$$\text{So } |\text{proj}_{\underline{b}} \underline{a}| = \frac{1}{2}$$

Each projection halves the length of the previous one.
 $(\frac{1}{2})^3 = \frac{1}{8}$ (A)

⑩ The proof states:

If a, b, c are 1 more than a multiple of 3,
 then abc is 1 more than a multiple of 3.

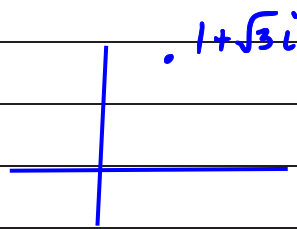
Contrapositive: If abc is not 1 more than a multiple of 3,
 then not all of a, b, c are 1 more than a
 multiple of 3. (D) fits this structure

Question 11

(a) (i) $z = 1 + \sqrt{3}i$
 $|z| = \sqrt{1^2 + (\sqrt{3})^2}$
 $= 2$

$\text{Arg}(z) = \tan^{-1}(\sqrt{3})$
 $= \frac{\pi}{3}$

$\therefore z = 2e^{i\frac{\pi}{3}}$



(ii) $zw = 2e^{i\frac{\pi}{3}} \times 3e^{i2\frac{\pi}{3}}$
 $= 6e^{i\pi}$
 $= -6$

(b) (i) $\frac{1}{2} \int \frac{2\sec^2 x}{1 + 2\tan x} dx = \frac{1}{2} \ln|1 + 2\tan x| + c$

(ii) $\int x e^x dx = \int x \times \frac{d}{dx}(e^x) dx$ (applying IBP)
 $= x e^x - \int e^x dx$
 $= x e^x - e^x + c$

(iii) $\int \cos^3 x dx = \int \cos x (1 - \sin^2 x) dx$ let $u = \sin x$
 $= \int (1 - u^2) du$ $du = \cos x dx$
 $= u - \frac{u^3}{3} + c$
 $= \sin x - \frac{1}{3} \sin^3 x + c$

(c) $z^2 - 2iz - 5 = 0$
 $(z - i)^2 = 5 + i^2$
 $(z - i)^2 = 4$

$z - i = \pm 2$
 $z = \pm 2 + i$

So $z = 2 + i$ or $z = -2 + i$

(finding Δ if using quadratic formula)

(d) False. If $x < 0$, then $y^2 < 0$ so y can't be real. ✓

$$\begin{aligned}(e) \quad \frac{z}{\bar{w}} &= \frac{1+5i}{3-2i} \\&= \frac{1+5i}{3+2i} \times \frac{3-2i}{3-2i} \quad \checkmark \\&= \frac{3+10+i(-2+15)}{9+4} \\&= \frac{13+13i}{13} \\&= 1+i \quad \checkmark, \quad \therefore a=1, b=1\end{aligned}$$

$$(f) \quad \vec{r} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+\lambda \\ -2+2\lambda \end{bmatrix}$$

$$\therefore \vec{OA} = \begin{bmatrix} 4+\lambda \\ -2+2\lambda \end{bmatrix}$$

$$\begin{aligned}\vec{OA} \cdot \vec{OB} &= \begin{bmatrix} 4+\lambda \\ -2+2\lambda \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\&= -4-\lambda-6+6\lambda \\&= -10+5\lambda\end{aligned}$$

$$\therefore \vec{OA} \cdot \vec{OB} = 0 \text{ when } \lambda = 2$$

$$\begin{aligned}\therefore \vec{OA} &= \begin{bmatrix} 4+2 \\ -2+2 \times 2 \end{bmatrix} \\&= \begin{bmatrix} 6 \\ 2 \end{bmatrix}\end{aligned}$$

$\therefore A$ is the point $(6, 2)$. ✓

✓ (correctly using dot product)

Question 12

$$\begin{aligned} \text{(a) (i)} \quad \underline{b} \cdot (\underline{a} - \underline{b}) &= \underline{b} \cdot \underline{a} - \underline{b} \cdot \underline{b} \\ &= \underline{b} \cdot \underline{a} - |\underline{b}|^2 \\ &= 9 - 3^2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad |\underline{a} - \underline{b}|^2 &= (\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b}) \\ &= \underline{a} \cdot \underline{a} - 2\underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b} \\ &= |\underline{a}|^2 - 2\underline{a} \cdot \underline{b} + |\underline{b}|^2 \\ &= 5^2 - 2 \times 9 + 9 \\ &= 16 \end{aligned}$$

$$\begin{aligned} \therefore |\underline{a} - \underline{b}| &= \sqrt{16} \\ &= 4 \end{aligned}$$

$$\text{(iii)} \quad \text{From (i), } \vec{OB} \perp \vec{BA}$$

$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \times |\vec{OB}| \times |\vec{BA}| \\ &= \frac{1}{2} \times |\underline{b}| \times |\underline{a} - \underline{b}| \\ &= \frac{1}{2} \times 3 \times 4 \\ &= 6 \text{ units}^2 \end{aligned}$$

$$\text{(b) (i)} \quad (\sqrt{x} - \sqrt{y})^2 \geq 0$$

$$\begin{aligned} \therefore x + y - 2\sqrt{xy} &\geq 0 \\ x + y &\geq 2\sqrt{xy} \\ \frac{x+y}{2} &\geq \sqrt{xy} \end{aligned}$$

Note: Squaring both sides of an inequality was penalised if proper justification wasn't given.

$$\begin{aligned} \text{(ii)} \quad \text{Using (i):} \quad 1 + a_1 &\geq 2\sqrt{a_1} \\ 1 + a_2 &\geq 2\sqrt{a_2} \\ 1 + a_3 &\geq 2\sqrt{a_3} \\ &\vdots \\ 1 + a_n &\geq 2\sqrt{a_n} \end{aligned}$$

(any correct application of AM-GM)

Multiplying:

$$\begin{aligned} (1+a_1)(1+a_2)(1+a_3)\dots(1+a_n) &\geq 2\sqrt{a_1} \times 2\sqrt{a_2} \times 2\sqrt{a_3} \times \dots \times 2\sqrt{a_n} \\ &\geq 2^n \sqrt{a_1 a_2 a_3 \dots a_n} \\ &\geq 2^n \text{ as } a_1 a_2 a_3 \dots a_n = 1 \end{aligned}$$

$$(c) \int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin\theta}$$

$$= \int_0^1 \frac{\frac{2dt}{1+t^2}}{1 + \frac{2t}{1+t^2}}$$

$$= \int_0^1 \frac{2dt}{1+t^2+2t}$$

$$= 2 \int_0^1 \frac{dt}{(1+t)^2}$$

$$= 2 \left[-(1+t)^{-1} \right]_0^1$$

$$= 2 \left[-\frac{1}{1+t} \right]_0^1$$

$$= 2 \left(-\frac{1}{1+1} + \frac{1}{1} \right)$$

$$= 2 \times \frac{1}{2}$$

$$= 1$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$$

$$= \frac{1}{2} (1+t^2)$$

$$\therefore d\theta = \frac{2dt}{1+t^2}$$

$$\theta=0, \quad t=0$$

$$\theta=\frac{\pi}{2}, \quad t=1$$

$$(d) I_n = \int_0^1 \frac{x^n}{x+1} dx, \quad n \geq 0$$

$$(i) I_n = \int_0^1 \frac{(x+1)x^{n-1} - x^{n-1}}{x+1} dx$$

$$= \int_0^1 \left(x^{n-1} - \frac{x^{n-1}}{x+1} \right) dx$$

$$= \left[\frac{x^n}{n} \right]_0^1 - I_{n-1}$$

$$= \frac{1}{n} - I_{n-1}$$

$$\begin{aligned} \text{(ii)} \quad I_0 &= \int_0^1 \frac{dx}{x+1} \\ &= [\ln(x+1)]_0^1 \\ &= \ln 2 - \ln 1 \\ &= \ln 2 \end{aligned} \quad \checkmark$$

$$\begin{aligned} \therefore I_1 &= \frac{1}{1} - \ln 2 \\ &= 1 - \ln 2 \end{aligned}$$

$$\begin{aligned} I_2 &= \frac{1}{2} - (1 - \ln 2) \\ &= \ln 2 - \frac{1}{2} \end{aligned} \quad \checkmark$$

Question 13

(a) (i) The coefficients of $P(z)$ are real, so complex zeroes occur in conjugate pairs.
 $\therefore$ the other zeroes are $a-ib$ and $b-ia$. ✓

(ii) Sum of zeroes:

$$(a+ib) + (a-ib) + (b+ia) + (b-ia) = 10$$

$$2a + 2b = 10$$

$$a + b = 5 \quad \textcircled{1}$$

Product of zeroes:

$$(a+ib)(a-ib)(b+ia)(b-ia) = 169$$

$$(a^2 + b^2)(b^2 + a^2) = 169$$

$$(a^2 + b^2)^2 = 169$$

$$a^2 + b^2 = 13, \text{ as } a, b \in \mathbb{R}$$

$$\textcircled{1}^2: a^2 + 2ab + b^2 = 25$$

$$2ab + 13 = 25$$

$$ab = 6$$

By inspection, as a & b are integers, $a=2, b=3$
(or $b=2, a=3$). ✓

(b) (i) $v = 2e^{-\frac{x}{2}}(x+1)^2$

$$v^2 = 4e^{-x}(x+1)^4$$

$$\frac{1}{2}v^2 = 2e^{-x}(x+1)^4$$

$$\frac{d}{dx} \left(\frac{1}{2}v^2 \right) = 2(-e^{-x}(x+1)^4 + e^{-x} \times 4(x+1)^3)$$

$$\therefore \ddot{x} = 2e^{-x}(x+1)^3[4 - (x+1)]$$

$$= 2e^{-x}(x+1)^3(3-x)$$

(ii) Note that $v > 0 \forall x$. Initially, $v = 2 \times e^0(0+1)^2 = 2 \text{ ms}^{-1}$, so particle always has positive displacement after $t=0$.
 $\ddot{x} = 0$ when $x=3$. ✓ Also, $\ddot{x} > 0$ for $0 < x < 3$ and $\ddot{x} < 0$ for $x > 3$. ✓ $\therefore$ Max. velocity occurs when $x=3$.

Note: $\ddot{x} = 0$ when $x=3$ is not sufficient justification, as it ignores the possibility of the object speeding up for $x > 3$ (i.e. v having an inflection at $x=3$).

(c) (i) $(\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta$ by De Moivre's theorem

$$\begin{aligned}(\cos\theta + i\sin\theta)^4 &= \binom{4}{0}\cos^4\theta + \binom{4}{1}\cos^3\theta(i\sin\theta) + \binom{4}{2}\cos^2\theta(i\sin\theta)^2 \\&\quad + \binom{4}{3}\cos\theta(i\sin\theta)^3 + \binom{4}{4}(i\sin\theta)^4 \\&= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta + i(4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta)\end{aligned}$$

Equating real: $\cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$

$$\begin{aligned}&= \cos^4\theta - 6\cos^2\theta(1 - \cos^2\theta) + (1 - \cos^2\theta)^2 \\&= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta \\&= 8\cos^4\theta - 8\cos^2\theta + 1\end{aligned}$$

(ii) $x^4 - 4x^2 + 1 = 0$

Let $x = 2\cos\theta$:

$$(2\cos\theta)^4 - 4 \times (2\cos\theta)^2 + 1 = 0$$

$$16\cos^4\theta - 16\cos^2\theta + 1 = 0$$

$$8\cos^4\theta - 8\cos^2\theta + \frac{1}{2} = 0$$

$$8\cos^4\theta - 8\cos^2\theta + 1 = \frac{1}{2}$$

$$\cos 4\theta = \frac{1}{2}$$

So the roots of the equation are $x = 2\cos\theta$, where $\cos 4\theta = \frac{1}{2}$

$$\cos 4\theta = \frac{1}{2}$$

$$4\theta = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 4\pi - \frac{\pi}{3} \quad (\text{taking further values gives duplicate results for } \cos\theta)$$
$$= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

$$\therefore \text{roots are } 2\cos\frac{\pi}{12}, 2\cos\frac{5\pi}{12}, 2\cos\frac{7\pi}{12}, 2\cos\frac{11\pi}{12}$$

(iii) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$$\begin{aligned}\therefore 4\cos^2\frac{\pi}{12} + 4\cos^2\frac{5\pi}{12} + 4\cos^2\frac{7\pi}{12} + 4\cos^2\frac{11\pi}{12} &= 0^2 - 2 \times -4 \\&= 8\end{aligned}$$

$$\therefore \cos^2\frac{\pi}{12} + \cos^2\frac{5\pi}{12} + \cos^2\frac{7\pi}{12} + \cos^2\frac{11\pi}{12} = 2$$

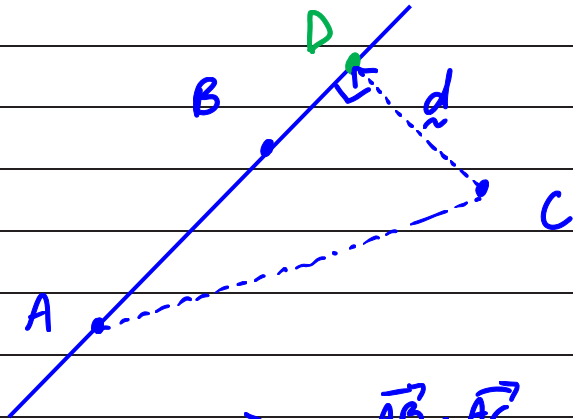
Question 14

(a) (i) $A(-2, 1, 7)$, $B(-1, 3, 5)$, $C(-4, 7, 3)$

$$\vec{AB} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad \vec{AC} = \begin{bmatrix} -2 \\ 6 \\ -4 \end{bmatrix}$$



(ii)



$$\text{Proj}_{\vec{AB}} \vec{AC} = \frac{\vec{AB} \cdot \vec{AC}}{\vec{AB} \cdot \vec{AB}} \vec{AB}$$

$$= \frac{-2 + 12 + 8}{1 + 4 + 4} \vec{AB}$$

$$= \frac{18}{9} \vec{AB}$$

$$= 2 \vec{AB}$$

$$= \begin{bmatrix} 2 \\ 4 \\ -4 \end{bmatrix}$$



$$\therefore \underline{d} = \vec{AD} - \vec{AC}$$

$$= \begin{bmatrix} 2 \\ 4 \\ -4 \end{bmatrix} - \begin{bmatrix} -2 \\ 6 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix}$$



$$\begin{aligned} \text{and } |d| &= \sqrt{4^2 + (-2)^2 + 0^2} \\ &= \sqrt{20} \\ &= 2\sqrt{5} \text{ units} \end{aligned}$$



$$(b) (i) \quad \frac{1}{p^2 - x^2} = \frac{A}{p+x} + \frac{B}{p-x}$$

$$1 = A(p-x) + B(p+x)$$

$$\underline{x=p:} \quad 1 = 2pB$$

$$B = \frac{1}{2p}$$

$$\underline{x=-p:} \quad 1 = 2pA$$

$$A = \frac{1}{2p}$$

$$(ii) \quad \int \frac{1}{p^2 - x^2} dx = \frac{1}{2p} \int \left(\frac{1}{p+x} + \frac{1}{p-x} \right) dx$$

$$= \frac{1}{2p} (\ln|p+x| - \ln|p-x|) + C$$

$$= \frac{1}{2p} \ln \left| \frac{p+x}{p-x} \right| + C$$

$$(c) (i) \quad m\ddot{y} = 10m - mkv^2$$

$$\ddot{y} = 10 - kv^2$$

Terminal velocity when $\ddot{y} = 0$:

$$kv^2 = 10$$

$$V = \sqrt{\frac{10}{k}}$$

, taking down as positive.

$$\ddot{y} = k \left(\frac{10}{k} - v^2 \right)$$

$$\frac{dv}{dt} = k(V^2 - v^2)$$

$$(ii) \quad \int \frac{dv}{V^2 - v^2} = \int k dt$$

$$\frac{1}{2V} \ln \left| \frac{V+v}{V-v} \right| = kt + C \quad \text{from (b)(ii)}$$

$$\ln \left| \frac{V+v}{V-v} \right| = 2Vkt + C_2$$

$$\left| \frac{V+v}{V-v} \right| = e^{2Vkt + C_2}$$

$$\frac{V+v}{V-v} = e^{C_2} \times e^{2Vkt} \quad \text{as } v < V$$

When $t=0$, $v=0$: $\frac{V+0}{V-0} = e^{C_2} \times 1$

$$e^{C_2} = 1$$

$$\therefore \frac{V+v}{V-v} = e^{2Vkt}$$

$$V+v = (V-v)e^{2Vkt}$$

$$v(1+e^{2Vkt}) = V(e^{2Vkt}-1)$$

$$v = V \frac{e^{2Vkt}-1}{e^{2Vkt}+1}$$

(iii) When $t=1$, $v = \frac{1}{2}V$:

$$\frac{1}{2}V = V \frac{e^{2Vk}-1}{e^{2Vk}+1}$$

$$e^{2Vk} + 1 = 2e^{2Vk} - 2$$

$$e^{2Vk} = 3$$

$$2Vk = \ln 3$$

but $V = \sqrt{\frac{10}{k}}$:

$$\sqrt{\frac{10}{k}} \times k = \frac{1}{2} \ln 3$$

$$\sqrt{10} \times \sqrt{k} = \frac{1}{2} \ln 3$$

$$k = \left(\frac{\ln 3}{2\sqrt{10}} \right)^2$$

$$= 0.03 \quad (2 \text{ decimal places})$$

Question 15

$$(a) \operatorname{Im}\left(\frac{z+\bar{z}}{z-\bar{z}}\right) \geq 1$$

Let $z = x+iy$:

$$\frac{z+\bar{z}}{z-\bar{z}} = \frac{x+iy + (x-iy)}{x+iy - (x-iy)}$$

$$= \frac{2x}{2iy} \times \frac{i}{i}$$

$$= \frac{ix}{-y}$$

$$\operatorname{Im}\left(-\frac{ix}{y}\right) \geq 1$$

$$-\frac{x}{y} \geq 1$$

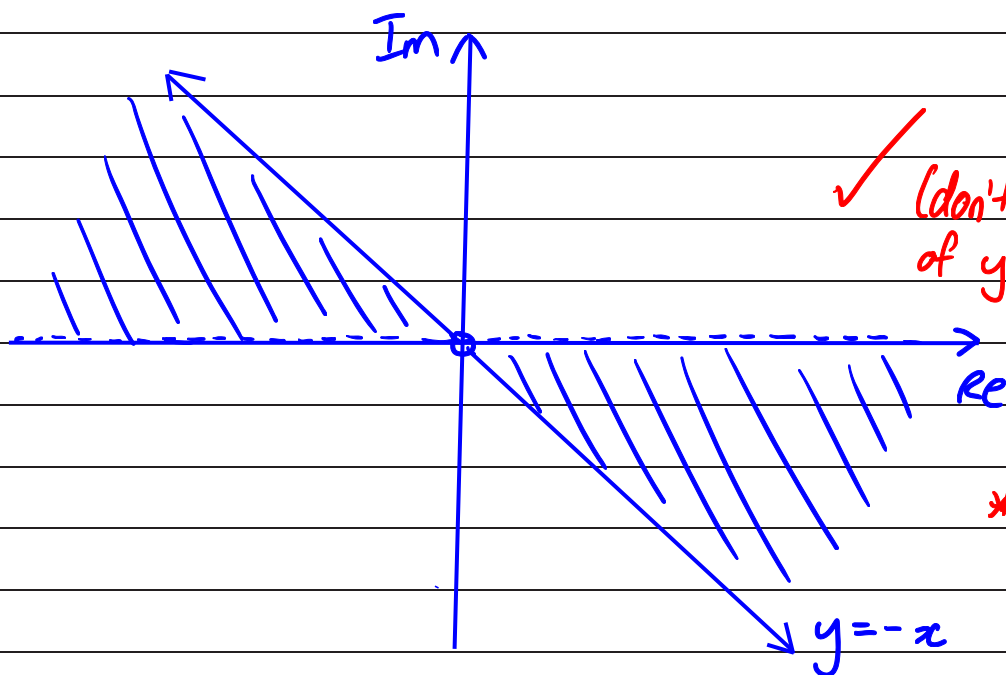
$$\frac{x}{y} \leq -1$$

$$xy \leq -y^2, \quad y \neq 0 \quad \checkmark \text{ (progress in dealing with inequality)}$$

$$y^2 + xy \leq 0$$

$$y(x+y) \leq 0$$

So either $y > 0$ and $x+y \leq 0 \rightarrow y \leq -x$
 or $y < 0$ and $x+y \geq 0 \rightarrow y \geq -x$



$\checkmark$ (don't penalise no indicator of $y=0$ not included)

* $\frac{x}{y} \leq -1$
 $\hookrightarrow y \leq -x$ + graph gets maximum 1 mark.

$$(b)(i) \quad z^7 = 1$$

$$\text{Let } z = \text{cis } \theta$$

$$\text{cis } 7\theta = \text{cis } 0$$

$$7\theta = 2k\pi$$

$$\theta = \frac{2k\pi}{7}, \quad k=0, \pm 1, \pm 2, \pm 3$$

✓ (valid method - diagram fine)

∴ the roots are

$$1, \text{cis } \frac{2\pi}{7}, \text{cis } (-\frac{2\pi}{7}), \text{cis } \frac{4\pi}{7}, \text{cis } (-\frac{4\pi}{7}), \text{cis } \frac{6\pi}{7}, \text{cis } (-\frac{6\pi}{7}).$$

$$(ii) \quad z^7 = 1$$

$$z^7 - 1 = 0$$

$$(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$$

As 1 is a zero of $z-1=0$,

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1$$

$$= (z - \text{cis } \frac{2\pi}{7})(z - \overline{\text{cis } \frac{2\pi}{7}})(z - \text{cis } \frac{4\pi}{7})(z - \overline{\text{cis } \frac{4\pi}{7}})(z - \text{cis } \frac{6\pi}{7})(z - \overline{\text{cis } \frac{6\pi}{7}})$$

$$= (z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)(z^2 - 2z \cos \frac{6\pi}{7} + 1)$$

(iii) Let $z=1$ in identity from part (ii):

$$1+1+1+1+1+1+1 = (2-2\cos \frac{2\pi}{7})(2-2\cos \frac{4\pi}{7})(2-2\cos \frac{6\pi}{7})$$

$$7 = 2^3 (1-\cos \frac{2\pi}{7})(1-\cos \frac{4\pi}{7})(1-\cos \frac{6\pi}{7})$$

$$7 = 2^6 \left(\frac{1}{2}(1-\cos \frac{2\pi}{7}) \right) \left(\frac{1}{2}(1-\cos \frac{4\pi}{7}) \right) \left(\frac{1}{2}(1-\cos \frac{6\pi}{7}) \right)$$

$$7 = 2^6 \sin^2 \frac{\pi}{7} \times \sin^2 \frac{2\pi}{7} \times \sin^2 \frac{3\pi}{7}$$

$$\therefore \sin^2 \frac{\pi}{7} \sin^2 \frac{2\pi}{7} \sin^2 \frac{3\pi}{7} = \frac{7}{2^6}$$

$$\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{2^3}$$

$$= \frac{\sqrt{7}}{8}$$

(c) $(2 - \frac{1}{n})^n > n$, $n \geq 2$

(A) Prove true for $n=2$:

$$\text{LHS} = (2 - \frac{1}{2})^2$$

$$= (\frac{3}{2})^2$$

$$= \frac{9}{4}$$

$$\text{RHS} = 2$$

$$< \frac{9}{4}$$

$\therefore$ true for $n=2$ ✓

(B) Assume true for some $n=k$, $k \geq 2$

i.e. $(2 - \frac{1}{k})^k > k$ (*)

Prove true for $n=k+1$:

RTP: $(2 - \frac{1}{k+1})^{k+1} > k+1$

$$\text{LHS} - \text{RHS} = (2 - \frac{1}{k+1})^{k+1} - (k+1)$$

$$> (2 - \frac{1}{k})^{k+1} - (k+1) \quad \text{as } k > 0$$

$$= (2 - \frac{1}{k})^k (2 - \frac{1}{k}) - (k+1)$$

$$> k(2 - \frac{1}{k}) - (k+1) \quad \text{from (*)} \quad \checkmark$$

$$= 2k - 1 - k - 1$$

$$= k - 2$$

$$\geq 0 \quad \text{as } k \geq 2 \quad \checkmark$$

$\therefore \text{LHS} - \text{RHS} > 0$ and $\text{LHS} > \text{RHS}$

From (A) and (B) above, statement holds for all integers $n \geq 2$ by mathematical induction.

(d)(i)

$$I = \int_a^b \frac{f(x)}{f(a+b-x) + f(x)} dx$$

$$\text{let } u = a+b-x$$

$$du = -dx$$

$$x=a, u=b$$

$$x=b, u=a$$

$$= \int_b^a \frac{f(a+b-u)}{f(u) + f(a+b-u)} -du$$

$$= \int_a^b \frac{f(a+b-u)}{f(u) + f(a+b-u)} du \quad \checkmark$$

$$= \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx \quad (\text{dummy variable})$$

$$\therefore 2I = \int_a^b \frac{f(x)}{f(a+b-x) + f(x)} dx + \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx$$

$$= \int_a^b \frac{f(x) + f(a+b-x)}{f(a+b-x) + f(x)} dx$$

$$= \int_a^b 1 dx$$

$$= b-a$$

$$I = \frac{b-a}{2} \quad \checkmark$$

(ii) $\int_1^3 \frac{x^2 dx}{\sqrt{10-x^2} + x}$

$$\text{let } u = x^2$$

$$du = 2x dx$$

$$x=1, u=1$$

$$x=3, u=9$$

$$= \frac{1}{2} \int_1^3 \frac{x \cdot 2x dx}{\sqrt{10-x^2} + x}$$

$$= \frac{1}{2} \int_1^9 \frac{\sqrt{u} du}{\sqrt{10-u} + \sqrt{u}} \quad \checkmark$$

$$= \frac{1}{2} \times \frac{9-1}{2} \quad \text{from (i)}$$

$$= 2 \quad \checkmark$$

Question 16

(a) (i) $\overline{z_3 z_2} = \overline{z_2 z_1}$

$$z_3 - z_2 = z_2 - z_1$$

$$2z_2 = z_3 + z_1$$

$$= z_2 z_1 + z_1$$

$$z_2(2 - z_1) = z_1$$

$$z_2 = \frac{z_1}{2 - z_1}$$



(ii) (α) Let $z_1 = x + iy$

$$z_2 = \frac{x + iy}{2 - x - iy} \times \frac{(2 - x) + iy}{(2 - x) + iy}$$

$$= \frac{x(2 - x) - y^2 + i(y(2 - x) + xy)}{(2 - x)^2 + y^2}$$



As z_2 is purely imaginary, $\text{Re}(z_2) = 0$

$$\therefore x(2 - x) - y^2 = 0$$

$$2x - x^2 - y^2 = 0$$

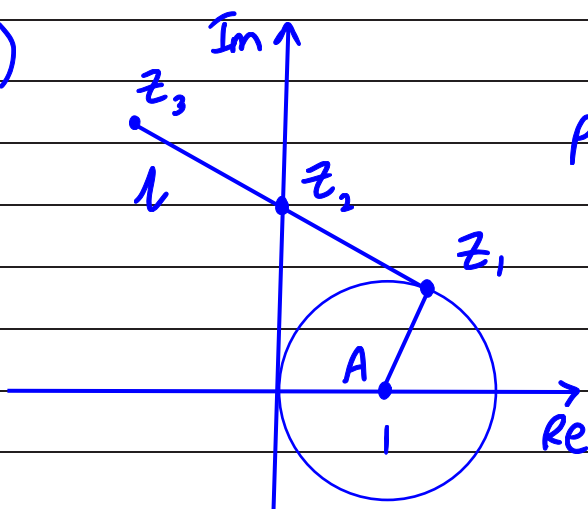
$$x^2 - 2x + y^2 = 0$$

$$(x - 1)^2 + y^2 = 1$$



which is a circle with centre $(1, 0)$ and radius 1.

(iii)



As z_1 lies on the circle, proving that $Az_1 \perp l$ is sufficient to show that l is a tangent.

$$\begin{aligned} \text{Arg}(z_3 - z_2) &= \text{Arg}(z_2 z_1 - z_2) \\ &= \text{Arg}(z_2(z_1 - 1)) \\ &= \text{Arg}(z_2) + \text{Arg}(z_1 - 1) \\ &= \pm \frac{\pi}{2} + \text{Arg}(z_1 - 1) \end{aligned}$$

(progress using valid method)

As the arguments differ by $\frac{\pi}{2}$, $Az_1 \perp l$, and l is a tangent.



(b) (i) For the vectors $\underline{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\underline{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$,

$$\underline{a} \cdot \underline{b} = a_1b_1 + a_2b_2 + a_3b_3$$

also, $\underline{a} \cdot \underline{b} = |\underline{a}| \times |\underline{b}| \times \cos\theta$, where θ is the angle between $\underline{a}$ and $\underline{b}$

$$-|\underline{a}| \times |\underline{b}| \leq \underline{a} \cdot \underline{b} \leq |\underline{a}| \times |\underline{b}|, \text{ as } -1 \leq \cos\theta \leq 1$$

$$\therefore |\underline{a} \cdot \underline{b}| \leq |\underline{a}| \times |\underline{b}|$$

$$(\underline{a} \cdot \underline{b})^2 \leq |\underline{a}|^2 \times |\underline{b}|^2, \text{ as LHS, RHS} \geq 0$$

$$\therefore (a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

$$(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \geq (a_1b_1 + a_2b_2 + a_3b_3)^2$$

(ii) $x^2 + 4y^2 + z^2 = x^2 + (2y)^2 + z^2$

Using part (i), let $a_1 = x, a_2 = 2y, a_3 = z$,

$$b_1 = 1, b_2 = \frac{1}{2}, b_3 = 1 :$$

$$\frac{(x^2 + (2y)^2 + z^2)(1^2 + (\frac{1}{2})^2 + 1^2)}{(x^2 + 4y^2 + z^2)(\frac{9}{4})} \geq (x \times 1 + 2y \times \frac{1}{2} + z \times 1)^2$$

$$x^2 + 4y^2 + z^2 \geq \frac{4}{9} \times 3^2 \text{ as } x+y+z=3$$

$$\therefore x^2 + 4y^2 + z^2 \geq 4$$

(c) (i) $\int_0^3 x f'(x) f(x) dx = \int_0^3 x f(x) \times \frac{d}{dx} (f(x)) dx$

$$= \left[x(f(x))^2 \right]_0^3 - \int_0^3 f(x)(x f'(x) + f(x)) dx$$

$$= 3(f(3))^2 - 0 - \int_0^3 (x f'(x) f(x) + (f(x))^2) dx$$

$$= 3 \times 5^2 - \int_0^3 x f'(x) f(x) dx - \int_0^3 (f(x))^2 dx$$

$$\therefore 2 \int_0^3 x f'(x) f(x) dx = 75 - 81$$

$$= -6$$

$$\therefore \int_0^3 x f'(x) f(x) dx = -3$$

$$\begin{aligned}
 & \text{or} \int_0^3 x f'(x) f(x) dx \quad \text{Let } u = x \quad v' = f'(x) f(x) \quad \checkmark \\
 & \quad \quad \quad u' = 1 \quad v = \frac{1}{2} (f(x))^2 \quad \checkmark \\
 & = \left[x \times \frac{1}{2} (f(x))^2 \right]_0^3 - \frac{1}{2} \int_0^3 (f(x))^2 dx \quad \checkmark \\
 & = \frac{3}{2} \times (f(3))^2 - 0 - \frac{1}{2} \times 81 \\
 & = \frac{3 \times 5^2 - 81}{2} \\
 & = -3 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad & (f'(x) f(x) + kx)^2 = (f'(x) f(x))^2 + 2k(x f'(x) f(x)) + k^2 x^2 \\
 \therefore & \int_0^3 (f'(x) f(x) + kx)^2 dx \\
 & = \int_0^3 (f'(x) f(x))^2 dx + 2k \int_0^3 x f'(x) f(x) dx + k^2 \int_0^3 x^2 dx \\
 & = \underset{\text{(given)}}{1} + \underset{\text{(from (i))}}{2k \times -3} + k^2 \left[\frac{1}{3} x^3 \right]_0^3 \\
 & = 1 - 6k + k^2 \left(\frac{1}{3} \times 3^3 - 0 \right) \\
 & = 1 - 6k + 9k^2 \quad \checkmark
 \end{aligned}$$

$$\therefore \int_0^3 (f'(x) f(x) + kx)^2 dx = (1 - 3k)^2$$

$$\text{Let } k = \frac{1}{3}: \int_0^3 \left(f'(x) f(x) + \frac{1}{3} x \right)^2 dx = 0$$

As $\left(f'(x) f(x) + \frac{1}{3} x \right)^2 \geq 0$, the only way the integral from 0 to 3 can be 0 is if $f'(x) f(x) + \frac{1}{3} x = 0$ ✓

$$f'(x)f(x) = -\frac{1}{3}x$$
$$\int f'(x)f(x)dx = \int -\frac{1}{3}x dx$$

$$\frac{[f(x)]^2}{2} = -\frac{1}{3} \times \frac{x^2}{2} + C$$

$$f(3) = 5: \quad \frac{5^2}{2} = -\frac{1}{3} \times \frac{3^2}{2} + C$$

$$C = \frac{5^2}{2} + \frac{3}{2}$$
$$= 14$$

$$\therefore \frac{[f(x)]^2}{2} = -\frac{1}{3} \times \frac{x^2}{2} + 14$$

$$[f(x)]^2 = 28 - \frac{1}{3}x^2$$

$$f(x) = \sqrt{28 - \frac{1}{3}x^2} \quad \text{as } f(x) > 0$$

$$\therefore f(0) = \sqrt{28}$$
$$= 2\sqrt{7}$$

